BUS63-3

Global Financial Systems & Economic Governance

Who governs the international economy, why systemic crises erupt, and when cooperation succeeds or fails

The capstone of the Economics arc: the full S1–S4 toolkit turned on the monetary system, the crisis library, and the institutions that govern — or fail to govern — globalisation.

Albert School — 22 hours

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Preface

This is the last course in the Economics arc, and it asks the arc's final question: **who governs the international economy?** You arrive with the whole toolkit. **BUS13-4 Microeconomics** gave you opportunity cost, surplus accounting, externalities, and the prisoner's dilemma. **BUS23-4 Macroeconomics** gave you the system — GDP, money creation, the policy mix, IS–LM, central banking, and debt/GDP dynamics. **BUS33-4 Labour Economics** gave you the distributional turn and the standard of causal proof. **BUS43-4 International Economics** opened the border: comparative advantage and its losers, the balance of payments, PPP and UIP, the Mundell–Fleming trilemma, and currency crises in three generations. This course assumes all of it. We will not re-derive the trilemma or re-explain a tariff; we **apply** them to the institutions and crises that make up the real international order.

The shift in altitude is the point. Every previous course studied an economy; this one studies the **governance** of the world economy — a domain with no world government, where order is produced (when it is produced at all) by a patchwork of institutions, treaties, and dominant powers, and where the central recurring fact is **cooperation failure**. Three questions organise the book. First, **how is the international monetary system structured**, and why do the imbalances it produces generate friction (Chapters 1–4)? Second, **why do systemic financial crises keep happening**, and what does each one teach about the limits of governance (Chapters 5–7)? Third, **what determines whether nations cooperate or defect** — on development, on climate, on the digital economy — and is there a structural reason cooperation is so hard (Chapters 8–10)?

Two integrating frames run through the whole book and deserve naming now. **Acemoglu and Robinson's** distinction between **inclusive** and **extractive** institutions is our lens for institutional **quality** — we use it to judge the IMF, the World Bank, the WTO, and the development record. **Rodrik's globalisation trilemma** — that deep economic integration, democratic politics, and national sovereignty cannot all three coexist — is our frame for cooperation **failure**, and it is where the book lands. As throughout the arc, each idea arrives as a concrete case before it is stated in the abstract, each model's limits are stated plainly, and the discipline is always the same: name the mechanism, identify the winners and losers, and say what the framework cannot settle.

1 The international monetary system: from gold to float

Before we can ask who governs the world economy, we need the stage on which governance happens: the **international monetary system** — the set of rules, conventions, and institutions that determine how exchange rates are set, how imbalances are settled, and how international liquidity is provided. It has been rebuilt twice in a century, and each design embodies a different answer to the **Mundell–Fleming trilemma** you met in **BUS43-4**.

1.1 The gold standard and its trilemma

> **Worked example — the discipline of gold.** Under the classical gold standard (roughly 1870–1914), Britain fixed the pound to a weight of gold, France fixed the franc, and so every currency was implicitly fixed to every other. If Britain ran a trade deficit, gold flowed out; the money supply contracted automatically; prices and wages fell; British goods became cheaper; the deficit corrected itself. No central bank **chose** this — the metal enforced it. The system delivered exactly two of the trilemma's three goods: **fixed exchange rates** and **free capital mobility**. The price was the third: **no monetary autonomy at all**. A country in recession could not ease; it had to let deflation do the adjusting, however brutal.

Definition 1 The **classical gold standard** fixed each currency to gold, and thereby to every other currency, with adjustment via automatic **gold flows** (the price–specie–flow mechanism). On the trilemma it chose the **fixed rate + open capital** edge, sacrificing **monetary autonomy**. Its virtue was **credibility and price stability** over the long run; its vice was the absence of any tool to fight recessions, so downturns were resolved by **deflation** — falling wages and output. It collapsed under the strain of World War I and could not be durably rebuilt in the 1920s.

Common pitfall Nostalgia for the gold standard ("sound money, no inflation") forgets **which corner of the trilemma it sacrifices**. Tying your hands to gold means you cannot respond to a banking panic or a demand collapse — the interwar gold standard is widely judged to have **deepened and propagated the Great Depression**, because countries that clung to gold longest suffered longest. There is no free corner. Every monetary system is a **choice about which goal to give up**, and the gold standard gives up the one you most need in a crisis.

1.2 Bretton Woods and the post-1973 float

Worked example — redesigning the system in 1944. Meeting at Bretton Woods in 1944, with the Depression and gold's rigidity fresh in mind, the architects (Keynes and White) wanted fixed rates **without** gold's deflationary straitjacket. Their compromise: fix each currency to the **US dollar**, fix the dollar to gold at \$35/ounce, but **restrict capital mobility** so that governments retained **monetary autonomy** to pursue full employment. On the trilemma, they moved to a different edge — **fixed rate + monetary autonomy** — by sacrificing **free capital movement**. The IMF and World Bank were created to manage it.

Definition 2 The **Bretton Woods system** (1944–1971) was a **dollar-exchange standard**: members pegged to the dollar, the dollar was convertible to gold at \$35/oz, and **capital controls** preserved national monetary autonomy. It made the **dollar** the world's reserve and settlement currency — a status that outlived the system itself. It broke down because of the **Triffin dilemma**: the world needed ever more dollars for liquidity, but the more dollars the US issued, the less credible its promise to redeem them for a fixed stock of gold. In 1971 the US suspended convertibility (the "Nixon shock"); by 1973 the major currencies **floated**.

Definition 3 The **post-1973 system** is a **managed float**: major currencies (dollar, euro, yen, pound) float against one another, while many smaller and emerging economies peg, crawl, or float within bands. The system chose the trilemma's **floating-rate + open-capital** edge for its core, restoring **monetary autonomy** by letting exchange rates move. The dollar retained — and even extended — its role as the dominant **reserve, invoicing, and safe-asset** currency, a primacy whose geopolitical uses are the subject of Chapter 4.

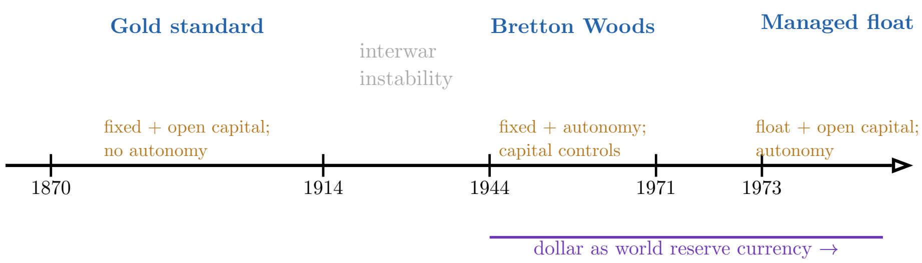

Figure 1: The international monetary system has been rebuilt twice, and each design sacrifices a **different corner of the Mundell–Fleming trilemma**. The gold standard kept fixed rates and open capital at the cost of all monetary autonomy; Bretton Woods bought autonomy back by imposing capital controls; the post-1973 float buys autonomy by letting exchange rates move. The **dollar's** role as the world's reserve currency, established at Bretton Woods, persisted through the transition to floating — the structural fact behind Chapter 4.

Remark Read the three systems as three **points on the same triangle**, not as progress from worse to better. Each was a rational response to the failure of its predecessor: the gold standard's rigidity bred Bretton Woods' controlled flexibility; Bretton Woods' Triffin contradiction bred the float. The trilemma is the deep structure; monetary history is the sequence of attempts to live with it.

2 Balance of payments and global imbalances

You built the **balance of payments** in **BUS43-4**: the current account (CA) and the financial account (FA) sum to zero, so a current-account deficit **is** a net capital inflow, and CA $= S - I$ ties the external balance to the gap between national saving and investment. We now use that machinery to read the defining structural feature of the contemporary system: large, **persistent** imbalances between surplus and deficit nations, and the political reasons they do not self-correct.

2.1 The identity, revisited as a governance problem

Worked example — the mirror of surpluses and deficits. In the 2000s–2010s, China and Germany ran current-account **surpluses** often exceeding 6–8% of GDP, while the United States ran persistent **deficits** of 3–5%. By the identity, every surplus is someone's deficit: the world's surpluses and deficits must sum (roughly) to zero. China and Germany were **net lenders** to the world — sending goods abroad and accumulating foreign assets (US Treasuries, in China's case) — while the US was the **net borrower**, absorbing the world's excess savings. These are not separate national stories; they are **one global configuration** seen from opposite ends.

Definition 4 A **global imbalance** is a large, persistent pattern of current-account surpluses in some countries mirrored by deficits in others. Because CA $= S - I$, a **surplus** country saves more than it invests at home (excess saving exported as capital), and a **deficit** country invests more than it saves (importing capital). The configuration is sustainable only as long as deficit countries can keep **borrowing** and surplus countries are willing to keep **lending**; a sudden unwillingness to finance the deficit is precisely a **sudden stop** — the trigger of many crises in the chapters ahead.

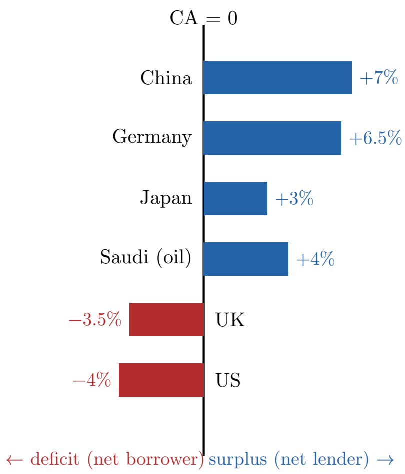

Figure 2: Global imbalances, schematic (current-account balance, % of GDP, mid-2010s orders of magnitude). **Surplus** economies — China, Germany, Japan, oil exporters — are **net lenders** to the world; **deficit** economies — the US, the UK — are **net borrowers**. The bars must (globally) net to zero: the world's deficits are financed by the world's surpluses. The persistence of this configuration, not its existence in any one year, is what makes it a governance problem.

2.2 Why imbalances persist: adjustment and its political blockages

Worked example — the adjustment that does not happen. Textbook adjustment is symmetric: a surplus country's currency should appreciate (or its wages and prices rise), eroding its competitiveness until the surplus shrinks; a deficit country's currency depreciates until its deficit shrinks. Why doesn't it? China for years **resisted** RMB appreciation by intervening (buying dollars, accumulating reserves) to protect its exporters. Germany, locked inside the euro, **cannot** let its currency rise relative to its euro-area partners at all — the single currency removes the adjustment channel entirely. The mechanism exists; **politics and institutions block it**.

Common pitfall Keynes saw the asymmetry in 1944 and lost the argument. The burden of adjustment falls almost entirely on **deficit** countries: a deficit country **must** eventually stop borrowing (markets force it), but a surplus country faces **no equivalent pressure** to stop lending — accumulating reserves is comfortable. Keynes proposed penalising **persistent surpluses** too; the US (a creditor in 1944) refused. The result is a system with a built-in **deflationary bias**: adjustment is forced on the weak, not shared. This asymmetry is a recurring theme — it returns in the euro crisis (Chapter 7), where surplus Germany faced no symmetric obligation to deficit Greece.

Remark Global imbalances are not **intrinsically** pathological. A young, fast-growing economy **should** run a deficit (importing capital to build its future); an ageing, high-saving economy **should** run a surplus (saving for retirement). The question, exactly as with a single country's deficit in **BUS43-4**, is **why** saving and investment diverge and whether the position is **sustainable**. The danger is not the imbalance itself but the **sudden reversal** of the capital flows that finance it — which is where systemic crises begin.

3 The Mundell–Fleming trilemma, applied comparatively

The trilemma was a **result** in **BUS43-4**; here it is an **analytical instrument**. The claim of this chapter is that you can characterise any major economy's entire monetary–exchange posture — and predict the costs it will pay — by identifying **which corner of the trilemma it has chosen to sacrifice**. We do this for the three systemically important cases: the Eurozone, the United States, and China.

3.1 The three corners, three regions

Definition 5 Recall the **trilemma**: a country cannot simultaneously have (i) a **fixed exchange rate**, (ii) **free capital mobility**, and (iii) **independent monetary policy**. It must sacrifice one. The three systemically important economies have each made a **different** choice, and each pays a characteristic price:

- the **Eurozone** sacrifices **national monetary autonomy** — one ECB rate for nineteen economies with open capital and irrevocably fixed intra-area rates;
- the **United States** sacrifices nothing it values, because as issuer of the reserve currency it floats **and** keeps autonomy **and** open capital — the trilemma binds it least;
- **China** sacrifices **free capital mobility** — capital controls let it manage the RMB **and** run an independent monetary policy.

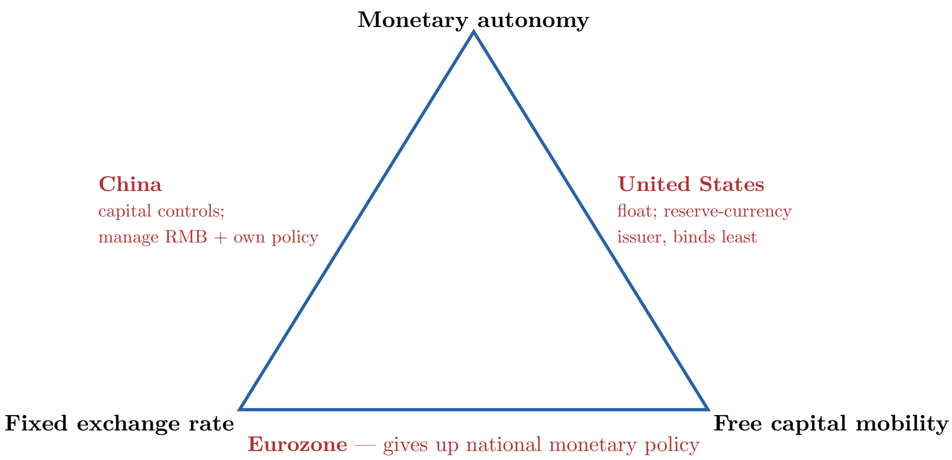

Figure 3: The Mundell–Fleming trilemma as a classifier of the three systemically important economies. Each occupies a **different edge**, having sacrificed the **opposite vertex**. The Eurozone surrenders national monetary autonomy to a single central bank; China closes its capital account to keep both a managed currency and its own interest-rate policy; the US, uniquely, escapes the bite of the trilemma because the world's demand for dollars lets it float with open capital and full autonomy at once.

3.2 The cost each region pays

> **Worked example — the Eurozone's missing shock absorber.** When the 2008 crisis hit, Spain and Ireland needed loose money and a weaker currency; Germany needed neither. But the ECB sets **one** rate, and the euro is **one** currency. Spain could not devalue to regain competitiveness, nor cut its own rates. With the exchange-rate and monetary shock absorbers both removed by the choice to give up autonomy, the **entire** adjustment fell on **internal devaluation** — years of wage cuts, austerity, and 25%+ unemployment. That is the price of the Eurozone's corner, and Chapter 7 shows how it nearly broke the union.

> **Worked example — America's exorbitant privilege.** Because the world wants to hold dollars — for trade invoicing, reserves, and safe assets — the US can borrow from the rest of the world **in its own currency**, at low rates, almost without limit. It runs large deficits (Chapter 2) without the financing crisis those deficits would trigger in any other country. Charles de Gaulle's minister called this the **"exorbitant privilege"**. The US sacrifices the **least** to the trilemma not because it is cleverer but because of the **structural** demand for its currency — a privilege Chapter 4 shows it has learned to wield as a weapon.

> **Common pitfall** China's corner is not free either — it is **expensive to maintain**. Capital controls require constant administrative effort, leak over time, and frustrate China's ambition to make the RMB a global reserve currency (which **requires** open capital markets — the very corner China sacrifices). China is thus caught in a slow dilemma: to internationalise the RMB it must open the capital account, but opening it forfeits either the managed exchange rate or monetary autonomy. The trilemma does not just describe China's present choice; it **bounds its future ambitions**.

4 Exchange rates as geopolitical instruments

So far the exchange rate has been an economic variable. This chapter treats it as an **instrument of power**. Once you see that the monetary system has a **centre** — the dollar — you can see why control of that centre confers leverage, and why the exchange rate has become a theatre of geopolitical conflict: currency manipulation, dollar primacy, and financial sanctions.

4.1 Currency manipulation and dollar primacy

Definition 6 **Currency manipulation** is deliberate official intervention to hold a currency **below** its market value — typically by buying foreign assets (accumulating reserves) — to subsidise exports and run a surplus. It is the exchange-rate face of the mercantilist surplus of Chapter 2. The difficulty is **evidential**: distinguishing manipulation from legitimate reserve management or monetary policy is genuinely hard, which is why "manipulation" accusations (the US against China for two decades) are as much political as technical.

Definition 7 **Dollar primacy** is the dollar's dominance as the world's **reserve** currency (≈60% of official reserves), **invoicing** currency (most commodities and much trade priced in dollars), and **safe asset** (US Treasuries). It gives the United States the **exorbitant privilege** of Chapter 3 — cheap borrowing in its own currency — and, crucially, **control of the plumbing**: because most cross-border dollar payments ultimately clear through US-linked systems, the US can **deny access** to them. Primacy is not just an economic convenience; it is the foundation of financial statecraft.

4.2 Financial sanctions: weaponising the network

Worked example — cutting Russia off, 2022. After the 2022 invasion of Ukraine, the US and allies **froze** roughly $300bn of the Russian central bank's foreign reserves and cut major Russian banks out of **SWIFT** (the messaging network for cross-border payments). Reserves a country **thought** it owned became unusable overnight; banks could not settle dollar or euro transactions. This is **interdependence weaponised**: the very integration that made Russia rich made it vulnerable. No army was involved — the weapon was access to the financial network the dollar sits at the centre of.

Common pitfall Financial sanctions are powerful but **self-eroding** — the same lesson as the optimal tariff in **BUS43-4**. Each use of dollar primacy as a weapon teaches targets (and watchful non-targets) to **build alternatives**: non-dollar payment rails, central-bank reserves diversified into gold and RMB, bilateral local-currency trade. The static coercive gain is real; the dynamic cost is that weaponisation **accelerates the fragmentation of the very system that makes the weapon work**. This tension — between using centrality and preserving it — is the strategic heart of monetary geopolitics, and a direct instance of the cooperation problem the book builds toward.

Remark Notice the throughline from Chapters 2–4: the surplus/deficit configuration, the US's trilemma escape, and the weaponisation of the dollar are **three faces of one fact** — the asymmetry of a system with a single dominant currency at its centre. Asymmetry creates privilege; privilege invites use; use erodes the asymmetry. Keep this in view: it is why "the international monetary system" is inseparable from "international power".

5 Minsky and the anatomy of a systemic crisis

We now turn to the second great question of the course: **why do systemic financial crises keep happening?** The orthodox models of **BUS23-4** mostly treated the financial system as a passive conduit for monetary policy. **Hyman Minsky** insisted the opposite: the financial system **generates its own instability**, endogenously, in good times. His **financial-instability hypothesis** is the spine of the crisis library in Chapter 7.

5.1 Stability breeds instability

> **Worked example — the boom that plants the bust.** A long economic expansion has no defaults; lenders and borrowers grow confident. A firm that once borrowed cautiously — comfortably covering interest **and** principal from cash flow — sees competitors lever up and prosper, and follows. Credit standards loosen because **nothing has gone wrong**. Asset prices rise, which makes the new debt look safe (the collateral is worth more), which justifies still more lending. The very **tranquillity** of the good times is what erodes the caution that produced it. Stability is **destabilising**: it breeds the leverage that the next shock will detonate.

Definition 8 Minsky's **financial-instability hypothesis** classifies borrowers by how they can service debt:

- **hedge** finance: cash flow covers both interest and principal — safe;
- **speculative** finance: cash flow covers interest but not principal, so debt must be **rolled over** — fragile to credit conditions;
- **Ponzi** finance: cash flow covers **neither**; the borrower relies entirely on **rising asset prices** to refinance — viable only while the bubble inflates.

The hypothesis: during long booms the financial system drifts **endogenously** from hedge toward speculative and Ponzi structures, so the system becomes **more fragile the longer stability lasts**. Crisis is not an external shock hitting a sound system; it is the **internal product** of the preceding calm.

5.2 The anatomy of a bubble

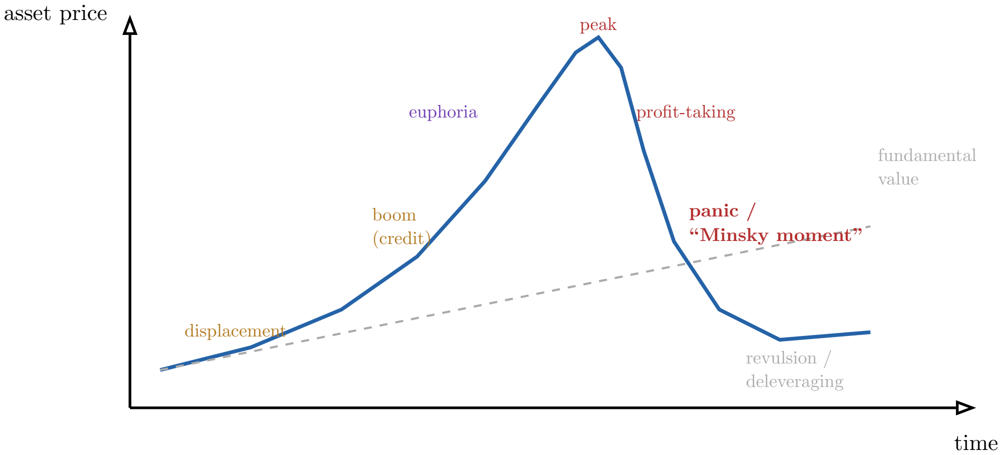

Figure 4: The anatomy of a bubble (Minsky–Kindleberger). A **displacement** (a new technology, deregulation, cheap credit) starts prices rising; a credit-fuelled **boom** lifts them above fundamental value; **euphoria** takes hold as Ponzi finance spreads and "this time is different" reasoning dominates. At the **peak** insiders take profits; when enough do, the **Minsky moment** arrives — a scramble for liquidity in which everyone deleverages at once, prices overshoot **below** fundamentals, and the crash feeds on itself. The curve is generated **endogenously**: the height of the boom builds the depth of the bust.

Definition 9 The **anatomy of a systemic crisis** runs in stages: a **displacement** shifts expectations; a **credit boom** and rising **leverage** amplify it; **euphoria** pushes prices above fundamentals; a **Minsky moment** — the point at which overstretched borrowers must sell to meet obligations — triggers a **fire sale**; **contagion** spreads the distress as falling asset prices and counterparty fears cascade through interconnected balance sheets. The three accelerants to watch are **leverage** (small price falls wipe out thin equity), **maturity/ liquidity mismatch** (short-term funding of long-term assets), and **interconnection** (one firm's liability is another's asset).

Common pitfall The seductive error in every bubble is **"this time is different"** — the belief that some structural change (the internet, securitisation, a new monetary regime) has abolished the old limits. Reinhart and Rogoff's history of financial crises is one long refutation: the **details** differ every time, but the **mechanism** — leverage built in calm, detonated by a shock — is monotonously the same. The analyst's discipline is to ask, at the height of any boom, **where are the Ponzi units, and what happens when refinancing stops?** — not to assemble reasons the music will not end.

6 The systemic crisis case library

Minsky gives the grammar; this chapter reads four sentences written in it. The **1997 Asian crisis**, the **2008 Global Financial Crisis**, the **2010–2015 European sovereign-debt crisis**, and the **2022 inflation shock** are not a random list — they are a **library** chosen so that each exposes a **different** failure of governance. Read them with the full toolkit: balance of payments, the trilemma, Minsky, and the IS–LM/policy-mix machinery of **BUS23-4**.

6.1 1997: the Asian crisis (a trilemma–balance-sheet crisis)

Worked example — the twin crisis of 1997. East Asian economies pegged to the dollar with **open** capital accounts (the trilemma corner that sacrifices monetary autonomy) and attracted huge **short-term dollar** inflows. Banks and firms borrowed in dollars but earned in baht, won, or rupiah — a **currency mismatch**. When confidence cracked and capital reversed (a **sudden stop**), the pegs broke; the local-currency value of the dollar debt **exploded**; banks and firms failed together — a **twin crisis** — and contagion leapt from Thailand to Korea to Indonesia. This is the **third-generation** currency crisis of **BUS43-4** and the **balance-sheet** fragility of Minsky, fused.

6.2 2008: the Global Financial Crisis (a shadow-banking crisis)

Worked example — how a housing slowdown broke the world. US mortgage lending loosened to **subprime** borrowers (Ponzi units: repayment required **house prices to keep rising**). Mortgages were **securitised** — bundled into mortgage-backed securities and CDOs, stamped AAA, and sold worldwide — so risk was **dispersed and disguised** across a lightly regulated **shadow banking** system (investment banks, money-market funds, repo) funded **short** and invested **long**. When house prices fell, the AAA tranches proved toxic; nobody knew who held the losses; short-term funding **froze**; Lehman failed; the panic became global through the **interconnection** Minsky warned of. A regional housing correction became a world depression-scare because **leverage, mismatch, and opacity** turned it systemic.

Definition 10 The 2008 build-up combined every accelerant: **leverage** (banks 30:1, thin equity), **securitisation** (risk repackaged until it was unreadable), **shadow banking** (maturity transformation **outside** the regulated, deposit-insured perimeter), and **global propagation** (European banks held US mortgage paper and funded in dollars). The policy response — central banks as **lender of last resort** at massive scale, **QE**, bank recapitalisation, fiscal stimulus, and later **Dodd–Frank/Basel III** re-regulation — is widely judged to have **prevented a second Great Depression**. The open question of **proportionality** is whether rescuing the system also rescued the **reckless** (moral hazard) and whether the **distributional** bill — bailouts for banks, austerity for households — seeded the political backlash of Chapter 10.

6.3 2010–2015: European sovereign debt (monetary union without fiscal union)

Worked example — a currency without a state. Greece, Ireland, Portugal, Spain, and Italy shared the euro but kept **national** budgets. Markets had long treated all euro-area sovereign debt as equally safe; when Greece's true deficit emerged in 2009, they abruptly **repriced** peripheral debt, and yields spiralled. Trapped in the trilemma's **give-up-autonomy** corner (Chapter 3), these countries **could not devalue or print** — the two escape valves a sovereign with its own currency has. A **self-fulfilling** doom loop set in: fears of default raised yields, which **caused** the insolvency feared, which bound fragile **banks** (holding sovereign bonds) to fragile **states** (backstopping banks).

Definition 11 The European sovereign-debt crisis was a crisis of **monetary union without fiscal union**: a single currency and central bank, but no common budget, no joint debt, and (initially) no lender of last resort for **sovereigns**. The **sovereign–bank doom loop** — weak banks weaken the state that guarantees them, and vice versa — was its signature mechanism. The repairs the euro area **did** make: Draghi's **"whatever it takes"** (2012) and **OMT**, which gave the ECB a credible sovereign-backstop and broke the self-fulfilling spiral; the **ESM** bailout fund; and a partial **banking union** (single supervision). The repairs it **did not** make: genuine **fiscal union** (joint budget, Eurobonds) and a full common deposit insurance — so the **fundamental** design flaw remains only **partly** fixed.

Common pitfall The euro crisis is the cleanest demonstration of Chapter 2′s **adjustment asymmetry**. Adjustment fell almost entirely on **deficit** periphery countries through brutal **internal devaluation** (austerity, wage cuts, mass unemployment), while **surplus** Germany faced no symmetric obligation to reflate. A monetary union that removes the exchange-rate shock absorber **needs** fiscal transfers to replace it; the euro area built a currency union without the transfer union, and the periphery paid the difference. This is the institutional gap Rodrik's trilemma (Chapter 9) predicts.

6.4 2022: the inflation shock (a policy-mix stress test)

Worked example — the return of inflation. Post-COVID **supply** shocks (disrupted chains, the Chapter 10 fragility realised) met massive pandemic **demand** support; the Ukraine war spiked **energy and food** prices. Inflation hit 8–11% across advanced economies — gone for a generation. The Fed and ECB, having held rates near zero, **tightened sharply** (the fastest hiking cycle in decades) to stop **second-round effects** — the wage–price spiral in which workers, expecting inflation, demand raises that **cause** more inflation (the de-anchoring of expectations from the Phillips-curve story of **BUS23-4**).

Definition 12 The 2022 episode is best read through the **policy-mix and IS–LM** framework of **BUS23-4**. A combined **supply** shock (cost-push, shifting the supply side) and **demand** overhang produced inflation that monetary tightening had to break by cooling demand — at the risk of recession (the Phillips-curve trade-off). **Fed/ECB coordination** mattered because uncoordinated tightening moves exchange rates (a sharp dollar appreciation exported inflation to everyone pricing imports in dollars). The **proportionality** question: was the tightening **late** (inflation dismissed as "transitory" too long) and then **overdone**, or a broadly successful engineering of **disinflation without a deep recession**? — a genuinely open verdict the framework lets you argue rather than assert.

The crisis library, in one view. Each case is a **different governance failure** read with the **same** toolkit. **1997**: a trilemma corner (peg + open capital) plus **balance-sheet** mismatch → twin crisis and contagion. **2008**: **leverage, securitisation, and shadow banking** turned a housing correction systemic — a pure Minsky build-up in a lightly governed financial system. **2010–15**: **monetary union without fiscal union** — a sovereign–bank doom loop with no devaluation escape, half-repaired. **2022**: a **supply-plus-demand** inflation shock testing the **policy-mix** and central-bank coordination. The common thread: crises are **endogenous** (Minsky), **propagate through interconnection**, and expose the precise **governance gap**

— a missing regulator, a missing fiscal union, a missing coordinator — that the institutions of the next chapters exist, imperfectly, to fill.

7 Institutions, institutional quality, and development

We now confront the question of **who** is supposed to govern, and **how well**. The postwar order built three institutions — the **IMF**, **World Bank**, and **WTO** — to manage money, development, and trade. To judge them, and to judge the wider record of **why some nations are rich and others poor**, we adopt **Acemoglu and Robinson's** distinction between **inclusive** and **extractive** institutions as our quality lens.

7.1 The institutional-quality lens

> **Worked example — Nogales, Arizona and Nogales, Sonora.** One city, split by a border fence. Same geography, same climate, same culture, the same name. North of the fence, incomes are several times higher, schooling longer, life safer; south of it, poorer and more violent. Geography and culture cannot explain the gap — they are identical. **Institutions** can: north of the fence, secure property rights, the rule of law, and political accountability; south, historically, the opposite. Acemoglu and Robinson's claim in one image: **it is institutions, not geography or culture, that decide whether a society prospers**.

Definition 13 **Acemoglu and Robinson** distinguish two institutional types:

- **inclusive** institutions secure broad property rights, enforce contracts impartially, allow wide participation, and reward innovation — they make it **worthwhile to invest and create**, generating **sustained** growth;
- **extractive** institutions concentrate power and funnel resources to a narrow elite; they can deliver **bursts** of growth (the elite can mobilise resources) but not **sustained** growth, because they suppress **creative destruction** — the competitive entry that threatens incumbents.

Their thesis: prosperity is **political** before it is economic. The lens generalises: we judge an **international** institution by whether its design is **inclusive** (broad voice, accountable, rule-bound) or **extractive** (captured by a few powers, imposing conditionality without legitimacy).

7.2 Judging the IMF, World Bank, and WTO

Definition 14 The three pillars of the postwar order, with mandate, instrument, and the **legitimate critique** each faces:

- the **IMF** — **crisis lender** and macro-surveillance body; instrument: emergency loans tied to **conditionality** (policy reforms required of borrowers). Critique: 1990s conditionality was often **pro-cyclical and ideological** (austerity that deepened the slump, as many argue it did in the 1997 Asian crisis), and its **governance** over-weights the US and Europe (the US holds an effective veto; the head is by convention European) — **extractive** on the A&R lens.
- the **World Bank** — **development lender** for long-term projects and poverty reduction. Critique: a mixed project record, past **one-size-fits-all** prescriptions, and the same governance asymmetry (its head is by convention American).

- the **WTO** — **trade-rules** body (Chapter 4 of **BUS43-4**): binds tariffs, enforces non-discrimination, adjudicates disputes. Critique: paralysed **Appellate Body** (since 2019), weak on **services, subsidies, and state capitalism**, and a stalled negotiating function.

Common pitfall Judging these institutions requires holding **two** truths at once, and partisans of each side drop one. They **are** governance-asymmetric — voting weights and leadership conventions entrench the powers that built them in 1944, which is a real **legitimacy** deficit on the inclusive/extractive lens. **And** they **do** provide genuine **global public goods** — a lender of last resort for countries, development finance, an escape from the tariff prisoner's dilemma — that no decentralised market produces. "Abolish them" and "they are beyond criticism" are both lazy. The analytic task is to ask, institution by institution, **whose voice is represented, whose conditionality is imposed, and whether the public good justifies the asymmetry**.

7.3 The development debate

Worked example — does foreign aid work?. A village has no bednets, so malaria keeps everyone too sick to work, so they stay too poor to buy bednets — a **poverty trap**. **Jeffrey Sachs** concludes: a **Big Push** of foreign aid can snap the trap, funding the bednets, fertiliser, and clinics that let growth begin. **William Easterly** retorts: decades of aid have little growth to show, aid props up **bad governments** (an extractive-institutions problem), and "planners" cannot engineer development from outside. Who is right? **Banerjee and Duflo** answer: stop arguing in the abstract — **test it**, bednet by bednet, with **randomised controlled trials** (the causal-inference standard of **BUS33-4**), and let evidence settle which specific interventions work.

Definition 15 The modern **development debate** arrays several positions:
- **Sachs** — **Big Push**: large coordinated aid breaks poverty traps;
- **Easterly** — aid **critique**: top-down aid fails and entrenches bad institutions; development comes from **home-grown** incentives and accountability;
- **Banerjee–Duflo** — the **RCT/experimental** turn: abandon grand theory, **measure** what works intervention by intervention;
- **Moyo** — aid as **harmful** to Africa, fostering dependency and corruption, crowding out trade and investment.

Behind the policy dispute sits Acemoglu–Robinson's deeper claim: **no** amount of aid substitutes for **inclusive institutions** — aid into an extractive polity leaks to the elite.

Definition 16 Two complementary widenings of the development lens:
- **Amartya Sen's capabilities approach**: development is not GDP but the expansion of people's real **freedoms and capabilities** — health, education, agency, the substantive ability to live a life one values. Income is a **means**, not the end — the same "GDP is not welfare" caution you met in **BUS13-4**, made into a theory of development.
- **Branko Milanovic** on **convergence and divergence**: globally, **between-country** inequality has **fallen** (Asia catching up — the elephant's hump of **BUS43-4**), even as

within-country inequality has often **risen**. Whether the world is converging or diverging depends entirely on **which inequality you measure** — and both are happening at once.

8 Global public goods, cooperation failure, and the globalisation trilemma

Here the book reaches its integrating idea. The recurring fact across the monetary system, the crisis library, and the institutions is that **cooperation keeps failing** even when it would benefit everyone. Why? **Rodrik's globalisation trilemma** gives the deepest answer, and it generalises the prisoner's dilemma of **BUS13-4** from firms to nations.

8.1 Global public goods and the free-rider problem

> **Worked example — the atmosphere nobody owns.** A stable climate is a **global public good**: everyone benefits from emissions cuts, and no country can be excluded from the benefit. So each country reasons: "global emissions depend negligibly on **my** costly cuts; let **others** cut and I will enjoy the stable climate for free." Every country reasons identically — so **nobody** cuts enough, and the good is undersupplied. This is the **free-rider problem**, and it is the prisoner's dilemma of **BUS13-4** played by 190 countries with **no world government** to enforce the cooperative outcome.

Definition 17 A **global public good** (climate stability, financial stability, pandemic preparedness, an open trading system) is **non-excludable** and **non-rival** at world scale. Because benefits accrue to all regardless of who pays, each nation has an incentive to **free-ride**, so the good is **systematically undersupplied**. The defining difficulty of global governance is that there is **no sovereign** above nations to compel contribution — cooperation must be **self-enforcing**, sustained only by repeated interaction, reciprocity, and institutions (Chapter 8) that are themselves weak. **Cooperation failure is the default**; cooperation is the thing that needs explaining.

8.2 Rodrik's globalisation trilemma

> **Worked example — the euro, the trilemma made concrete.** Recall the euro crisis (Chapter 7). Deep economic integration (one currency, free capital) collided with **national democracy**: Greek voters wanted to reject austerity, but the integrated system's creditors and rules constrained what any national government could deliver. Either the rules bind and democracy is hollowed ("there is no alternative"), or democracy asserts itself and the integration fractures (Grexit threats). You **cannot** have full economic integration, full national sovereignty, **and** full democracy at once — the euro forced the choice in miniature.

Definition 18 **Rodrik's globalisation trilemma**: a society cannot simultaneously have all three of (i) **deep economic integration** (hyper-globalisation), (ii) the **nation-state** (national sovereignty), and (iii) **democratic politics**. At most two:

- integration + nation-state, **sacrificing democracy** — the "golden straitjacket", where global market discipline dictates national policy regardless of voters (the classical gold standard, or austerity-era periphery);

- integration + democracy, **sacrificing the nation-state** — **global federalism**, pushing democracy up to a supranational level (the EU's unfinished, contested direction);
- nation-state + democracy, **sacrificing deep integration** — **managed globalisation**, where nations democratically choose how much integration to accept (the Bretton Woods compromise, and Rodrik's own preference).

The trilemma explains **why** cooperation fails: pushing globalisation **too deep** forces nations to choose between democracy and sovereignty, and democratic publics eventually **revolt** — the backlash of the elephant-curve trough (**BUS43-4**).

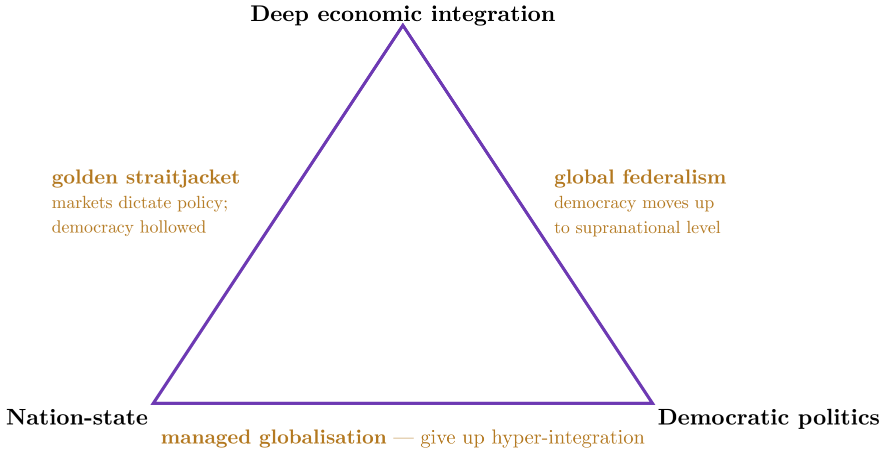

Figure 5: Rodrik's **globalisation trilemma** — the structural diagnosis of cooperation failure, and the integrating frame of the course. Each vertex is a goal; each edge is a feasible combination that sacrifices the opposite vertex. Push integration to the top and a nation must give up **either** democracy (the golden straitjacket, where global markets overrule voters) **or** sovereignty (global federalism). The only way to keep **both** the nation-state and democracy is to accept **less** integration — Rodrik's "managed globalisation", the Bretton Woods compromise. The backlash against hyper-globalisation is the trilemma asserting itself.

Common pitfall Do not confuse Rodrik's trilemma with Mundell–Fleming's (Chapter 3). Mundell–Fleming is a **monetary** constraint (fixed rate, capital mobility, monetary autonomy); Rodrik's is a **political-economy** constraint (integration, sovereignty, democracy) that **contains** it — the loss of monetary autonomy inside the euro is one **channel** by which deep integration hollows national democracy. The course's two triangles are nested: the monetary trilemma is **how** the deeper political trilemma bites on the exchange rate. Seeing the link is the synthesis the whole arc was building toward.

9 Governing the frontier: climate and the digital economy

The book closes on the two governance challenges where cooperation failure is being fought out **right now**: the **climate transition** and the **digital/AI economy**. Each is a global-public-good problem (Chapter 9) meeting Rodrik's trilemma, and each is a live test of whether the institutions of Chapter 8 can be rebuilt or extended.

9.1 Climate: carbon leakage and the CBAM

> **Worked example — why one country's carbon price leaks.** The EU prices carbon; a US or Chinese steelmaker does not. So EU steel becomes dearer, buyers switch to cheaper

unpriced imports, and EU production (with its emissions) simply **relocates** abroad — global emissions barely fall while EU industry shrinks. This is **carbon leakage**: the free-rider problem of Chapter 9 realised through trade. Unilateral climate ambition is **self-undermining** unless the carbon price is enforced **at the border** — which is exactly what the EU's **Carbon Border Adjustment Mechanism** (CBAM, the Pigouvian border levy you met in **BUS43-4**) attempts.

Definition 19 **Climate governance** is the hardest global-public-good problem: diffuse benefits, concentrated costs, a long horizon, and acute **distributional** conflict between rich historical emitters and poor current developers. **Carbon leakage** — emissions migrating to unpriced jurisdictions — undermines unilateral action; the **CBAM** is a **partial** response, levelling the field at the border and pricing the externality on imports. Its limits are real: it covers only some sectors, strains **WTO** compatibility, and developing exporters denounce it as **green protectionism** that burdens **their** development — the inclusive-vs-extractive and convergence debates of Chapter 8, fought on the carbon frontier.

9.2 Digital and AI governance: the fragmentation race

Worked example — three internets, three rulebooks. The same AI model faces three regimes. The **EU** regulates it as a **rights and safety** problem (GDPR, the AI Act) — precaution first. The **US** leans on **market-led innovation** with lighter federal rules — speed first. **China** treats the digital sphere as an instrument of **state control and industrial policy** — sovereignty first. There is no global rulebook; there are **three**, mutually incompatible. A firm operating worldwide must comply with all three at once, and the standards themselves become a **terrain of geopolitical competition**.

Definition 20 **Digital and AI governance** is fragmenting along a **US–EU–China** split — market-led, rights-led, and state-led models that do not interoperate. The live disputes: **platform taxation** (where is value created when a digital service has users in one country and profits booked in another? — the OECD's halting attempt at a global minimum tax is the cooperative response); the **standards race** (whoever sets the technical and regulatory standards for AI, 5G/6G, and data shapes the global market); and **data sovereignty**. As with climate, the structural problem is **Rodrik's**: deep digital integration collides with **national** democratic choices about privacy, speech, and security, so the world fragments into blocs rather than converging on common rules.

Common pitfall Resist the symmetry of pessimism **and** of techno-optimism. Cooperation failure is the **default** (Chapter 9), but it is not a law: the **Montreal Protocol** (ozone) and the post-2008 **Basel III** coordination show that, under the right conditions — concentrated and identifiable costs of **in**action, a small enough number of pivotal players, credible monitoring, and side-payments to compensate losers — cooperation **does** succeed. The analyst's job on any governance challenge is precisely to **diagnose those conditions**: name the public good, identify the free-riders, locate the distributional losers who must be compensated, and assess whether an institution exists (or could be built) to make cooperation **self-enforcing**. That diagnosis — not a verdict of "globalisation is doomed" or "markets will sort it out" — is the capstone skill this entire arc has been training.

The thread of this book. There is no world government, so international order is produced by institutions and dominant powers, and its default failure mode is **non-cooperation**. The **monetary system** (Ch. 1–4) is a sequence of answers to the Mundell–Fleming trilemma — gold, Bretton Woods, float — leaving a dollar-centred order whose **asymmetry** yields global **imbalances**, an American **privilege**, and a weaponisable dollar. **Systemic crises** (Ch. 5–7) are **endogenous** (Minsky: stability breeds instability), propagating through **leverage, mismatch, and interconnection**; the case library — 1997, 2008, 2010–15, 2022 — shows each crisis exposing a **specific** governance gap (a missing regulator, a missing fiscal union, a missing coordinator). **Governance** (Ch. 8–10) is judged by **Acemoglu–Robinson's** inclusive/extractive lens — applied to the IMF, World Bank, and WTO and to the **development debate** (Sachs–Easterly–Banerjee/Duflo–Moyo, Sen, Milanovic) — and its deep limit is **Rodrik's trilemma**: deep integration, the nation-state, and democracy cannot all coexist, so cooperation fails wherever globalisation is pushed past what democratic sovereignty will bear. Climate (CBAM, leakage) and digital/AI governance are this drama playing out now. The discipline throughout: **name the mechanism, identify the winners and losers, diagnose the conditions for cooperation, and state what the framework cannot settle.**

Exercises

1. Place the gold standard, Bretton Woods, and the post-1973 managed float on the Mundell–Fleming triangle, stating which corner each sacrifices and the characteristic cost of that choice. Explain the **Triffin dilemma** that brought down Bretton Woods, and why nostalgia for the gold standard underestimates the cost of its sacrificed corner.

2. Using $\mathrm{CA} = -\mathrm{FA} = S - I$, explain why China's surplus and the US deficit are "one configuration seen from two ends". Explain the **adjustment asymmetry** Keynes identified in 1944 — why the burden falls on deficit rather than surplus countries — and connect it to the euro crisis of Chapter 7.

3. Apply the Mundell–Fleming trilemma comparatively to the Eurozone, the US, and China: state which corner each sacrifices, name the price each pays, and explain why the US "binds least" (the exorbitant privilege) while China's chosen corner **bounds its ambition** to internationalise the RMB.

4. Explain **dollar primacy** and how it underpins **financial sanctions**, using the 2022 freezing of Russian reserves and SWIFT exclusion. Then explain, with the optimal-tariff analogy of **BUS43-4**, why weaponising the dollar is **self-eroding**.

5. State Minsky's **financial-instability hypothesis**, defining hedge, speculative, and Ponzi finance, and explain the claim that "stability breeds instability". Sketch the anatomy of a bubble through its stages (displacement → euphoria → Minsky moment → revulsion) and name the three accelerants that turn a local shock systemic.

6. Diagnose the **2008 GFC**: explain how leverage, securitisation, and shadow banking turned a US housing slowdown into a global crisis, and evaluate the **proportionality** of the policy response (lender of last resort, QE, re-regulation) against the moral-hazard and distributional objections.

7. Analyse the **2010–2015 European sovereign-debt crisis** as "monetary union without fiscal union". Explain the **sovereign–bank doom loop**, why periphery countries could not devalue or print, and identify which institutional repairs the euro area **did** make (OMT, ESM, banking union) and which it **did not** (fiscal union, Eurobonds).

8. Evaluate the **2022 inflation shock** using the **BUS23-4** policy-mix framework: distinguish the supply and demand drivers, explain why second-round effects justified sharp tightening, why Fed/ECB coordination mattered for exchange rates, and argue both sides of the proportionality question (late vs. overdone vs. soft landing).
9. Using **Acemoglu and Robinson's** inclusive/extractive lens, evaluate the IMF, World Bank, and WTO — mandate, instrument, and one legitimate critique each. Explain the "two truths" an honest assessment must hold together: governance asymmetry **and** genuine global-public-good provision.
10. Compare **Sachs** (Big Push), **Easterly** (aid critique), **Banerjee–Duflo** (RCTs), and **Moyo** on development and foreign aid, and relate the debate to Acemoglu–Robinson's institutional claim. Then explain how **Sen's** capabilities approach and **Milanovic's** convergence/divergence data each widen "development" beyond GDP.
11. State **Rodrik's globalisation trilemma** and draw its triangle, labelling the three feasible combinations (golden straitjacket, global federalism, managed globalisation) and the goal each sacrifices. Explain precisely how it **differs from** and **contains** the Mundell–Fleming trilemma, using the euro as the worked example of both.
12. Choose **one** governance challenge — climate (carbon leakage and CBAM) or digital/AI regulation (US–EU–China fragmentation, platform taxation, the standards race). Identify the global public good, name the free-riders and the distributional losers, explain the **structural** reason cooperation fails (using Chapters 9–10), and state the conditions under which it might nonetheless succeed.